



CIVIL MAINT. Department
SIMHADRI

Ref: SMTTP/Civil Maint/2021-22/LMI/816829

Date:

24/02/2022(13:14:29)

Subject: Review of LMI Inspection and Maintenance of Structures/Buildings in the Vicinity of Corrosive Chemicals, Effluents and Fumes

The LMI of "Inspection and Maintenance of Structures/Buildings in the Vicinity of Corrosive Chemicals, Effluents and Fumes" with Ref. LMI/OGN/OPS/CIV/012, Rev. No.2 was prepared and approved in the month of Jan'2020.

The LMI has been reviewed as on date and no changes are envisaged.

Put up for approval please.

Submitted by:

पांडब दत्ता - वरिष्ठ प्रबंधक

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On: 24/02/2022(13:14:29) **Ref:** SMTTP/Civil Maint/2021-22/LMI/816829

The LMI of "Inspection and Maintenance of Structures/Buildings in the Vicinity of Corrosive Chemicals, Effluents and Fumes" with Ref. LMI/OGN/OPS/CIV/012, Rev. No.3 may please be approved.

Submitted by:

जयंती गोपी किरण - उप महाप्रबंधक

Jayanthi Gopi Kiran - 008259 (DY. GENERAL MANAGER ,CIVIL MAINT.)
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On:25/02/2022(09:50:11) **Ref:**SMTTP/Civil Maint/2021-22/LMI/816829

LMI may please be approved.

Submitted by:

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On:25/02/2022(13:39:05) **Ref:**SMTTP/Civil Maint/2021-22/LMI/816829

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On:25/02/2022(13:43:11) **Ref:**SMTTP/Civil Maint/2021-22/LMI/816829

Please review entries on page no.2 of the revised document (given in annexure) as discussed.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:26/02/2022(13:42:37) **Ref:**SMTTP/Civil Maint/2021-22/LMI/816829

Page No.2 modified as discussed.

Submitted for approval please.

Submitted by:

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On:26/02/2022(18:30:49) **Ref:**SMTTP/Civil Maint/2021-22/LMI/816829

May please be accorded approval.

Submitted by:

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On:27/02/2022(13:06:15) **Ref:**SMTTP/Civil Maint/2021-22/LMI/816829

Submitted by:

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On:28/02/2022(09:27:27) **Ref:**SMTPP/Civil Maint/2021-22/LMI/816829

Approved & Submitted by:

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On:28/02/2022(16:42:33) **Ref:**SMTPP/Civil Maint/2021-22/LMI/816829

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/CIV/012

ISSUE NO.1; REVISION NO.:03; FEB'2022

**Inspection and Maintenance of Structures /
Buildings in the Vicinity of Corrosive Chemicals,
Effluents and Fumes**

Approved for
Implementation

CGM, SIMHADRI

Date :

LMI APPROVAL DOCUMENTS

1. TITLE OF LMI	Inspection and Maintenance of Structures / Buildings in the Vicinity of Corrosive Chemicals, Effluents and Fumes
2. a) DOCUMENT CODE b) REVISION/ISSUE NO.&DATE	LMI/OGN/OPS/CIV/012 ISSUE NO.1; REVISION NO.: 03; FEB'2022
3. ORIGINAL/REF.DOCUMENT	LMI/ONG/OPS/CIV/012 Issue No.: 1, Rev. 02, Jan'2020
4. PREPARED BY	Name: Shri. Pandab Datta Sr. Mgr. (O&M-Civil)
5. CHECKED BY	Name: Shri. J. Gopikiran DGM (O&M-Civil)
6. REVIEWED BY	Name: Shri. A. K. Mukhopadhyay AGM (MM-OS) Name: Shri. M. E. Kovoov GM (Maintenance) Name: Shri. G. C. Choukse GM (O&M)
7. APPROVED BY	Name: Shri. Diwakar Kaushik CGM (Simhadri)
8. DATE OF ISSUE	
9. INDEX LIST	Enclosed

NTPC	LMI	SIMHADRI																																													
TITLE: Inspection and Maintenance of Structures / Buildings in the Vicinity of Corrosive Chemicals, Effluents and Fumes REV: 3.00 DATE: FEB'2022 REF NO. LMI/OGN/OPS/CIV/012 PAGE: 01 OF 09																																															
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<u>2.0 INTRODUCTION:</u> The acid/alkaline handling areas in SIMHADRI power project. Like DM plant building neutralization pits, storage area of lime, alum and acids sodium chlorides and battery rooms are lined with acid resistant tiles, bit mastic coatings, and epoxy plants as barrier in between concrete surfaces and chemicals. The concrete surfaces if contracted with the acids and chlorides will be affected/corroded. There by the reinforcement bars will be exposed and the reinforcement too will be corroded, ultimately the structure will be collapse. The fumes of the acids also affect the windows (steel) plastering of walls and structural steel members. For smooth and uninterrupted power generation the safety of the structures and buildings, acid handling areas are to be safe and intact and this can be achieved by periodical inspection and marinating the acid/alkali proof lining in good condition. The objective of this LMI is to instruct and monitor the measures to be taken for inspection of the buildings and the structures in the vicinity of the acids and alkaline handling area. <u>3.0 SCOPE:</u> The scope of this LMI covers the inspection and maintenance of various protection linings and coatings at NTPC/SIMHADRI. The various are like DM plant, Effluent trenches, regeneration area, area around equipments, foundations, neutralization, pit, chemical house, chemical stores area, chemical lab area, alum solution preparation tank, acid, alkali store area, battery rooms, salt storing area, platforms, tanks, pump houses, ETP, CW chlorination BLDG, oil spillage area like oil cannels, oil storage rooms, cot, dot etc., CPU, structures on clarifiers, gravity filters and in the of NDCT etc. <u>4.0 REFERENCE DOCUMENT:</u> Operation guidance note no. REF no. LMI/OGN/OPS/CIV/012 circulated by operation services-NTPC corporate center IS2395, IS1475 <u>5.0 SUPERSEDED DOCUMENT:</u> None		

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TYPE OF LINING	POSSIBILITY TO CAUSE FAILURE	INSPECTION	REMEDIAL MEASURES	RESPONSIBILITY
1.0 Brick/Tile lining				
1.1 Jointing Mortar	Loose band, disintegration along corrosive chemicals to penetrate	1.Frequency of inspection every year except for alum, solution tanks where in its once in 3 months 2.Surface cleaning, visual inspection of joints, lining for cracks, desalting, and magnification or handle microscope 3.Lining for hollowness of the bending 4.Painting surfaces shall be inspected once in every six months 5.If required concrete samples to be taken and tested in chem. lab for chloride etc.	1.Through washing cleaning, chiseling, refill with resin mortar of same quality 2.Replaced cracked brick/tile 3.Injection grouting for depending of lining 4.Relining with epoxy mortar as per requirement but maintain at least 0.5m water level from bottom of basket to reduce impact 5.For concrete surface follow recommendation of IS 2395(P-II) CL 4.3 given Annex-I 6. For steel surfaces follow IS 1477(P-II) CL.7 given Annex-II	1.All inspection of exceptional reports (Annx-III) by the committee 2. Testing by chemistry dept. 3. Execution of repair/replacement by civil maint.
1.2 Bedding Mortar	Loose bond in bedding mortar, fall in vertical lining			
1.3 Bricks/Tiles	Cracks, defective material, accidental damage allowing seepage of harmful chemicals			
2.0 Epoxy Lining	Cold joints, pinholes cracks structural deformation chemicals weather etc.			
3.0 Bitumen lining				
4.0 Epoxy painting, chemical resistant	Blistering due to entrapped air, chemically effect, cracks, peeling, rusting			

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<u>7.0 METHODOLOGY OF PAINTING:</u>						
<u>RECOMMENDED PRATICE FOR MAINTAINANCE PAINTING ON CONCRETE AND PLASTER SURFACES</u>						
CONDITION OF SURFACE					CLEANING AND PAINTING PRATICE	
Blisteri ng And Softeni ng	Efflore - scence	Chalking	Cracking	Flaki ng		
Nil	Nil	Yes	Nil	Nil	Clean, rub down with abrasive paper, apply one finishing coat.	
Nil	Yes	Yes	Nil	Nil	Rub down the affected surface to bare plaster, allow to dry, touch with primer coat and then apply the required number of finishing coats and followed by a reviver coat on the entire surface, if required.	
Yes	Nil	Nil	Nil	Nil	Scrape off the soft film, allow the surface to dry completely, touch up with primer coat and then apply the required number of finishing coats and followed by a reviver coat on the entire surface, if required.	
Nil	Nil	Nil	Yes (Localised)	Nil	Rub down and even up the surface with sand paper to remove cracked film. touch up with primer, and apply the required number of finishing coats and followed by a reviver coat on the surface, if required.	
Nil	Nil	Nil	Nil	Yes	Remove the paint from the affected portion completely and follow part or full schedule of painting as the case may be based upon OEM's recommendations.	
Note:	In case of any serious characteristic failure mentioned above (except chalking), the paint should be completely removed and the full schedule of painting be followed.					

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RECOMMENDED MAINTENANCE PAINTING ON METAL SURFACES						
CONDITION OF SURFACE					CLEANING AND PAINTING PROCEDURE	
SL.NO	RUSTING	CRACKING	CHECKING	CHALKING		
i)	Nil	Nil	Nil	Yes	Clean rub with sandpaper; apply one reviver coat.	
ii)	Nil	Nil	Yes fine checking	Nil	Flat down to remove checked film; clean; apply one or two reviver coats	
iii)	Nil	Yes not reached the surface below	-	Nil	Treat as in A.2.5.1.2	
iv)	Yes	Nil	Nil	Yes	Treat as in A.2.5.1.5	
v)	a) Localised	Nil	Nil	Yes	Treat as in A.2.5.1.5	
	b) Localised	Nil	Yes	Nil	Treat as in A.2.5.1.5	
	c) Localised	-	-	-	Completely remove paint to bare metal and paint with full schedule	
vi)	Heavy Rusting	-	-	-	Completely remove paint to bare metal and paint with full schedule	

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<u>8.0 EXCEPTION REPORTING:</u> The exceptions encountered during inspection are to be reported as per the following format ----- Inspection committee. <table border="1"> <thead> <tr> <th>UNIT NO</th> <th>LOCATION</th> <th>EXCEPTION</th> <th>Remarks</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table>			UNIT NO	LOCATION	EXCEPTION	Remarks				
UNIT NO	LOCATION	EXCEPTION	Remarks							
<u>9.0 RESPONSIBILITY:</u> The responsibility of carrying out different tasks in relation to structures buildings and lining works in the vicinity of corrosive chemicals as detailed in the LMI are with the inspection committee consisting of representatives O&M (civil maintenance) and chemistry.										
<u>10.0 RECORD KEEPING:</u> The records to be maintained under the LMI include a) Compliance record of the activates detailed under this LMI b) Reports of inspection carried out c) Exception reporting The above records are to be maintained by MTP, MMD & O&M-Civil for relevant responsibilities.										
<u>11.0 REVIEW:</u> This document shall be reviewed by HOP (Simhadri) on two yearly basis.										

[illegible]

NTPC	LMI		SIMHADRI
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DISTRIBUTION LIST			
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COMPLIANCE RECORD		
STRUCTURE/CONCRETE STRUCTURE	ACTIVITY	COMPLIANCE